

Sibibalan Jeevanandam

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EDUCATION:

Ph.D., Mechanical Engineering, Purdue University – West Lafayette

Expected Completion: **May 2027**

Specialization: Systems, Measurements, and Control

Advisor: Dr. Neera Jain

B.Tech., Mechanical Engineering, Indian Institute of Technology (IIT)
Tirupati, India

GPA: 9.78/10, August 2018 – May 2022

RESEARCH INTERESTS:

Control-oriented Dynamic Modeling; State Estimation; Human-Machine Interaction

RELEVANT COURSEWORK:

Lumped System Theory, System Identification, Digital Signal Processing, Human Factors in Engineering

HONORS & AWARDS:

1. ME Winkelman Fellowship 2022
2. Summer Undergraduate Research Fellowship (SURF) Graduate Mentor Award, Purdue University, 2023

PUBLICATIONS:

Peer Reviewed Conference Papers

- P. Lankireddy, **S. Jeevanandam**, A. Choudhary, P. C. Deshmukh, K. Roberts, and S. R. Valluri, 'Solar cells, Lambert W and the LogWright functions', in *2023 International Conference on Electrical, Computer and Energy Technologies (ICECET)*, 2023, pp. 1–6.
- A. Maignan, L. Prabhat Reddy, **S. Jeevanandam** et al., 'The electronic properties of graphene nanoribbons and the offset logarithm function', *Materials Today: Proceedings*, vol. 54, pp. 7–13, 2022.

RESEARCH EXPERIENCE:

Graduate Research Assistant, Purdue University – West Lafayette

Aug 2022 – Present

- Co-developed a novel experiment design using System Identification principles to perturb cognitive factors such as trust in conditionally automated driving contexts.
- Co-developed an experimental testbed using Unreal Engine 5 to simulate conditionally automated driving (SAE Level 3) in a broad range of scenarios.
- Collected data from human subject experiments and used statistical tools to evaluate the experiment design.

Research Internship for Young Academics (RIYA), Ohio State University

May 2021-July 2021

- Explored the dynamics of a vibration absorber system with focus on non-linear paths using the multi-term harmonic balance method.

Undergraduate Researcher, IIT Tirupati, India

Dec 2019-May 2022

- Modeled a non-linear compliant rolling-sliding contact mechanism (cam-follower) when subjected to a period motion input using analytical and finite-element based methods.
- Studied the system's non-linear dynamic response in the frequency domain and evaluated the effect of the follower's compliance.

LEADERSHIP:

SURF 2023 Undergraduate Research Student Mentor, Purdue University

May 2023 – Aug 2023

- Mentored a student on developing an experimental platform using Unreal Engine 5 to serve as a testbed for collecting data to model human cognitive states during conditionally automated driving.

ICON Student Research Conference Presentation Co-Chair, Purdue University

Dec 2023 – Present

- Coordinating a presentation session for the 2024 Purdue ICON student research conference for students the chance to present their research and explore new research avenues.

SERVICE:

EPICS, Purdue University and IIT Tirupati

Sep 2019 – Dec 2020

- Worked in a team towards developing a Mobile Science Laboratory in rural schools in and around Tirupati to enhance the scientific temperament of students lacking access to laboratories.